Partner
Ecosystem
Success

SAP® PartnerEdge®

RAG with SAP HANA Vector Engine and Generative AI Hub

December, 2024

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SAP HANA Cloud Vector Engine

Vector

A vector is a list of numerical values (typically floats) that represents data in a multi-dimensional space. In the context of Generative AI, vectors are used to represent objects (e.g., a book, car, or customer record) by encoding their attributes or characteristics. Vectors enable comparisons, such as finding - similar objects based on their vector representations.

‘How to find the shortest path’ → *[0.245, 0.543, 0.467, 0.326, …..]*

Vector

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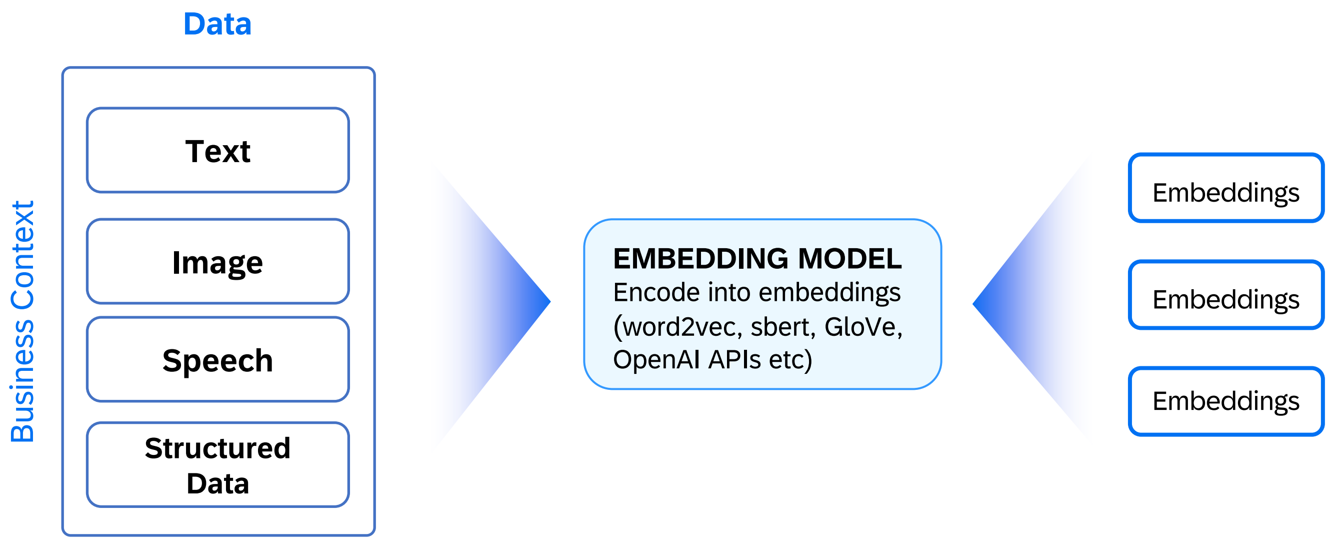
Embedding Model

A type of machine learning model that is designed to transform high-dimensional categorical data into lower-dimensional, continuous vector representations. This model is particularly used for transforming high-dimensional and complex data into a more concise and meaningful representation.

Example : Word2Vec, GloVe, Sentence Bert, Open AI APIs etc.



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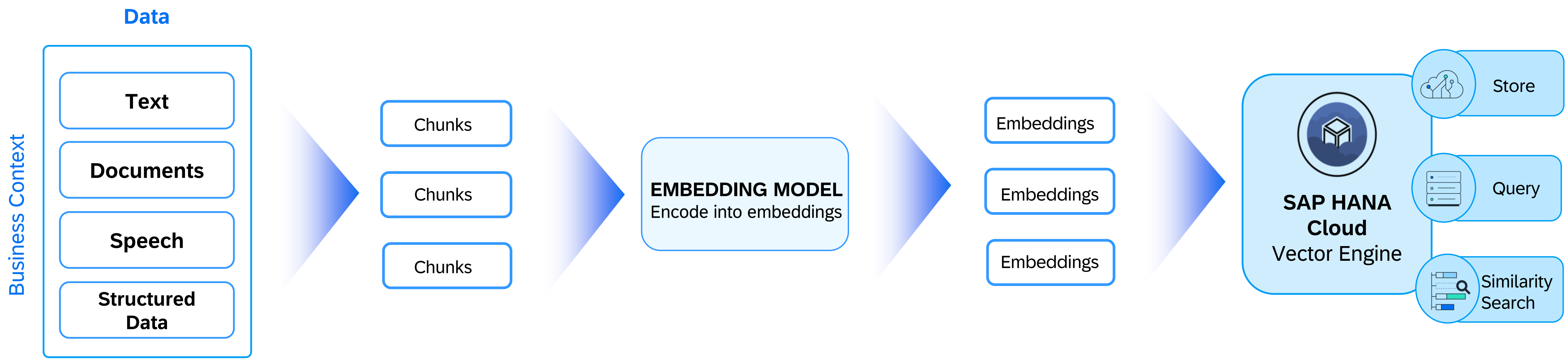

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Vector data type : REAL_VECTOR, TO_REAL_VECTOR

- Natively store high dimensional vectors
- E.g. text embeddings

Vector functions: L2DISTANCE(), COSINE_SIMILARITY()

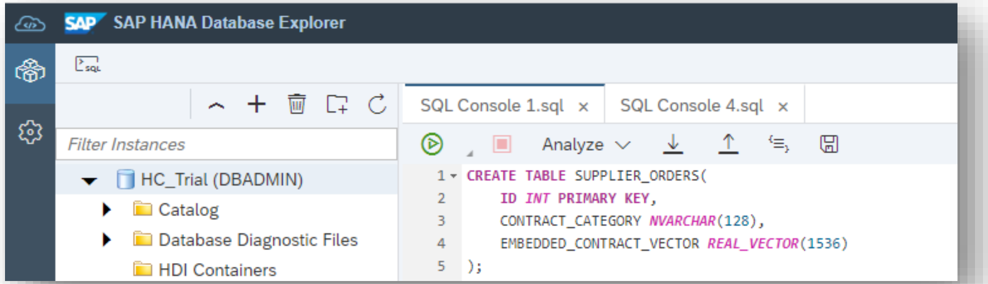
- Vector distance functions
- Combine vector functions with other SQL operations

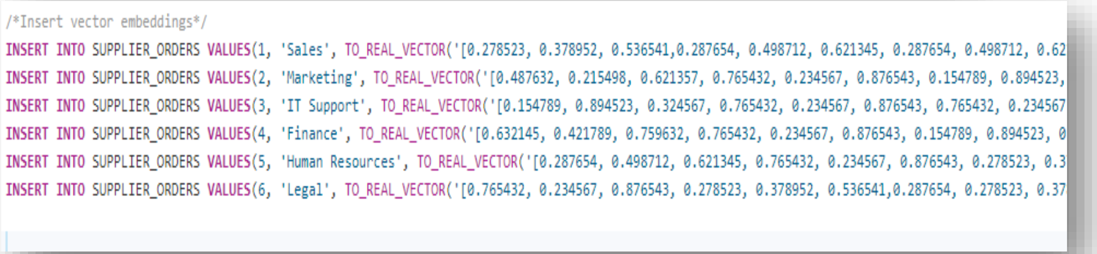
Consumption

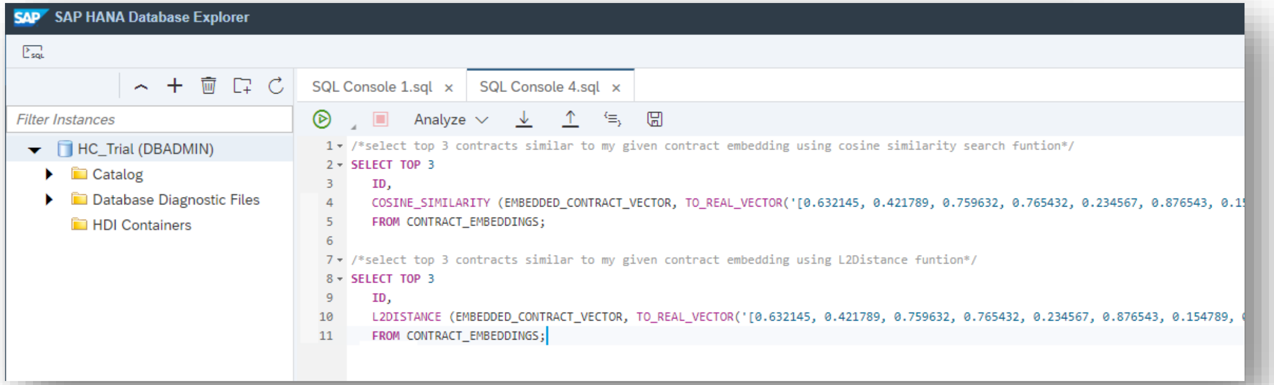
- SQL
- python (hana-ml, LangChain)

Roadmap & Vision

- In-database text embedding (VECTOR_EMBEDDING)
- Batch embedding (PAL_TEXTEMBEDDING)







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```
CREATE TABLE VECTORTAB (ID INT, V REAL_VECTOR(3));
```

Table Name

VECTORTAB

Columns | Indexes | Properties

	Name	SQL Data Type
1	ID	INTEGER
2	V	REAL_VECTOR

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REAL_VECTOR Data Type

Consists of REAL elements
(IEEE 754 single-precision floating-point).

The dimension can range from 1 to 65,000.

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```sql
SELECT TO_REAL_VECTOR('[17,19,23]') FROM DUMMY;

```

Result × Messages × History

Rows (1) SQL

	TO_REAL_VECTOR('[17,19,23]')
1	0300000000008841000098410000B841

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TO_REAL_VECTOR Function

Constructs a vector from a textual or binary representation, or from an array.

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```sql
SELECT TOP 20 *, L2DISTANCE(V, TO_REAL_VECTOR('[27,41,3]')) as L2_Distance
FROM VECTORTAB
ORDER BY L2_Distance;
```

Result × Messages × History

Rows (5)

	ID	V	L2_DISTANCE
1	5	030000000000D841000024420000544	0
2	2	030000000000C041000020420000544	3.1622776601683795
3	4	030000000000803F000030410000044	44.45222154178574
4	3	030000000000E0400000803F0000504	60
5	1	0300000000000040000040400000A04	66.12866246946176

L2DISTANCE Function

Computes L2 Distance of two vectors

The result is a DOUBLE number greater or equal to 0.

The greater the result, the greater the distance between the vectors.

The smaller the result, the smaller the distance between the vectors.

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```
SELECT TOP 20 *, COSINE_SIMILARITY(V, TO_REAL_VECTOR('[27,41,53]')) as cos_similarity
FROM VECTORTAB
ORDER BY cos_similarity DESC;
```

Result × Messages × History

Rows (5) SQL

	ID	V		COS_SIMILARITY
1	5	030000000000D8410000244200005442		1
2	2	030000000000C0410000204200005442		0.9992827614405838
3	1	03000000000000400000404000000A040		0.9925144139971539
4	4	030000000000803F0000304100000442		0.8858383888812046
5	3	030000000000E0400000803F00005041		0.8596067271179668

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COSINE_SIMILARITY Function

Computes the Cosine Similarity of two vectors

The result is a DOUBLE number between -1 and 1.

The greater the result, the more similar are the vectors.

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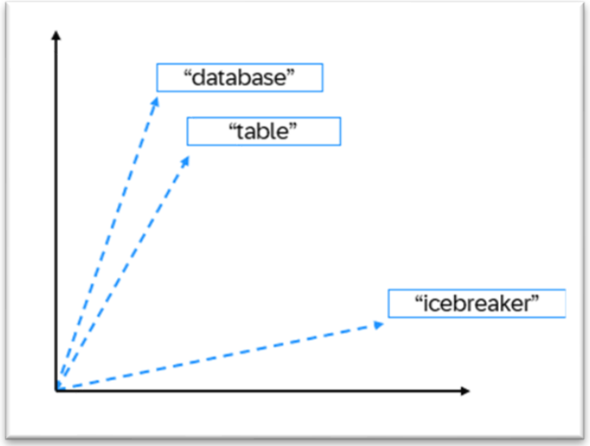
Similarity Search Example

Search Term: “Database”

Search Term (attributes stored as Vector Embeddings): [0.18, -0.55, 0.77, ……]

Compare against table of records stored in the database:



Most similar to ‘Database’

Least similar to ‘Database’

Term	Vector Embeddings	Distance	Rank
Table	[0.18, -0.65, 0.75, ……]	0.24	1
Schema	[0.12, 0.75, 0.14, …….]	0.28	2
Token	[0.67, 0.45, -0.16, ……]	0.57	3
Icebreaker	[0.89, 0.77, 0.34, …….]	0.68	4

Considering distance calculated using L2Distance() function

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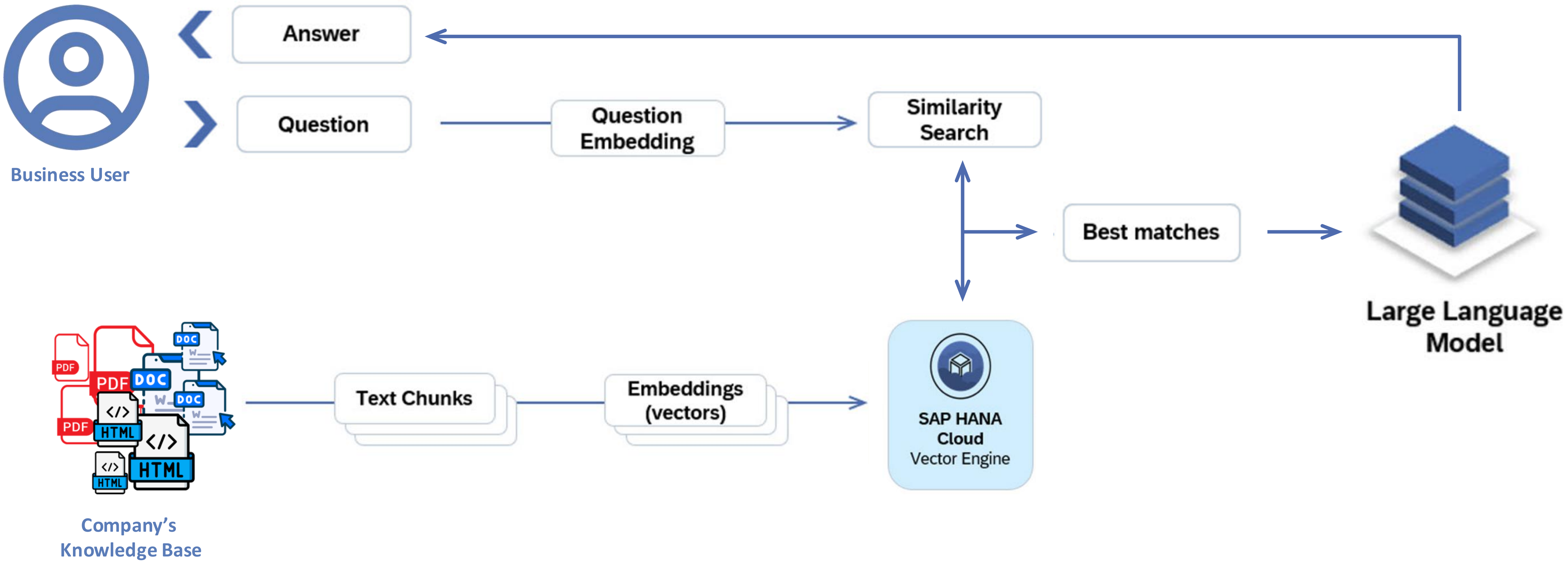
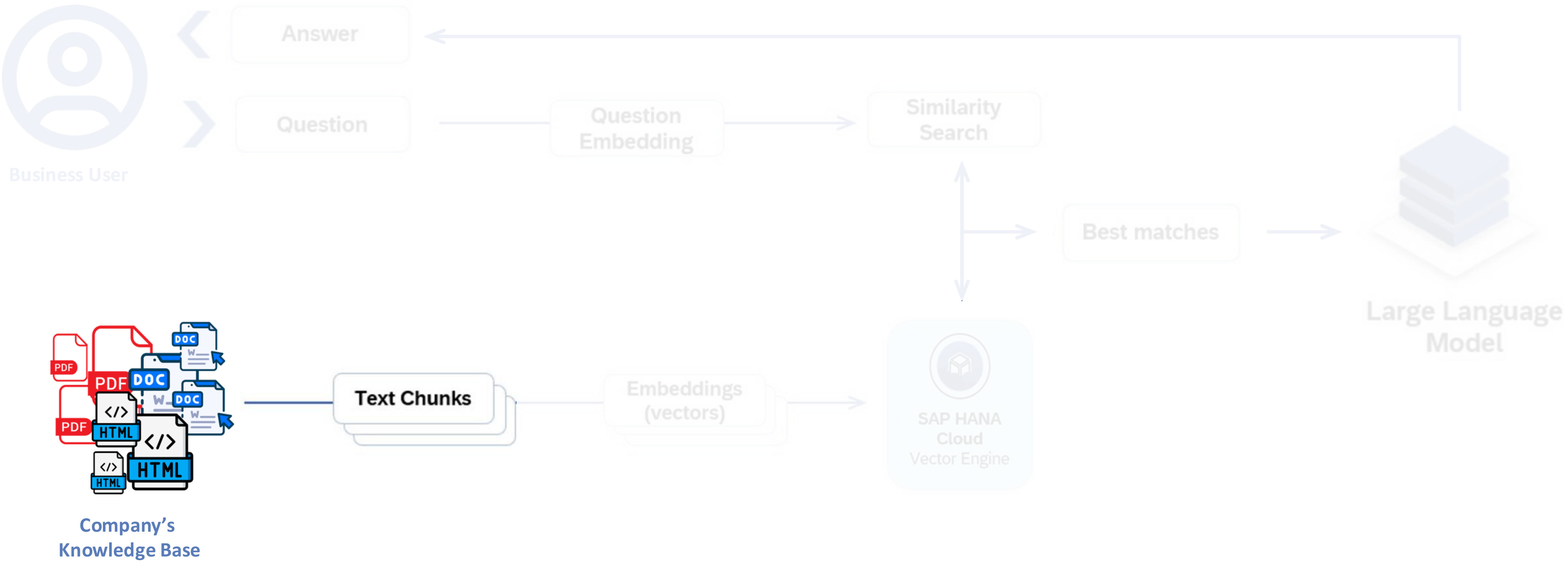
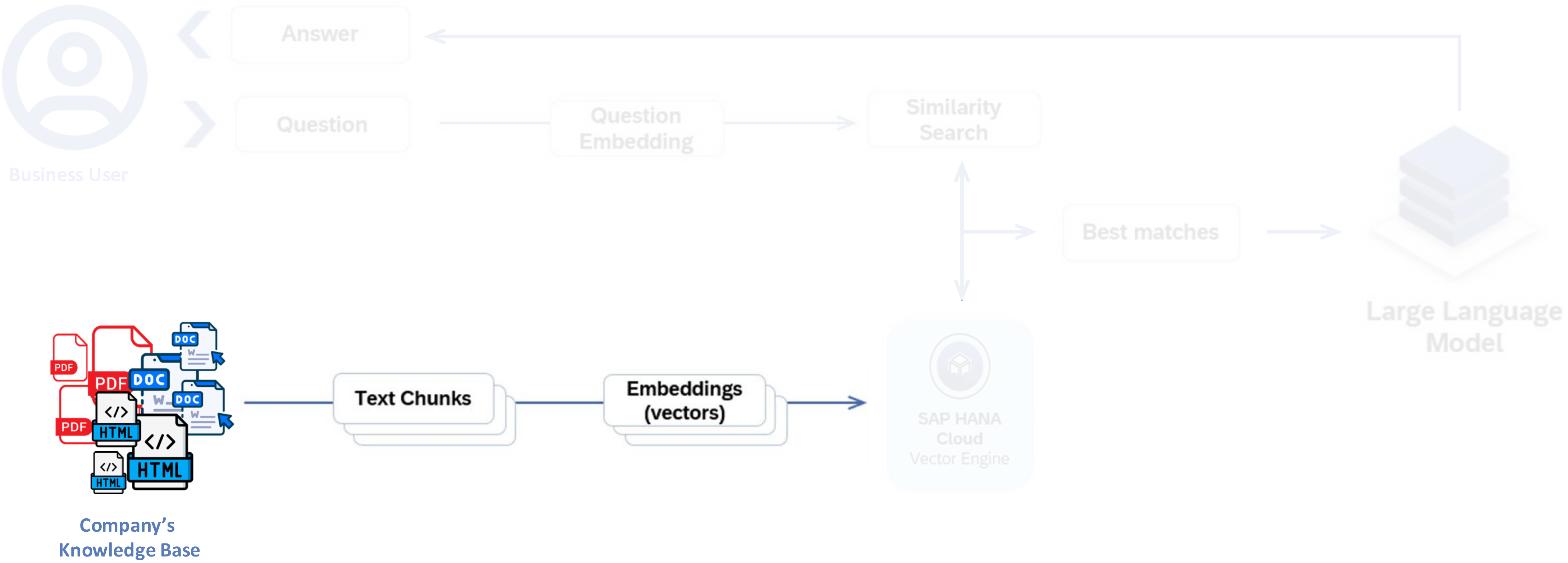


Split the documents into text chunks



Split the documents into text chunks

Generate the Embeddings from texts

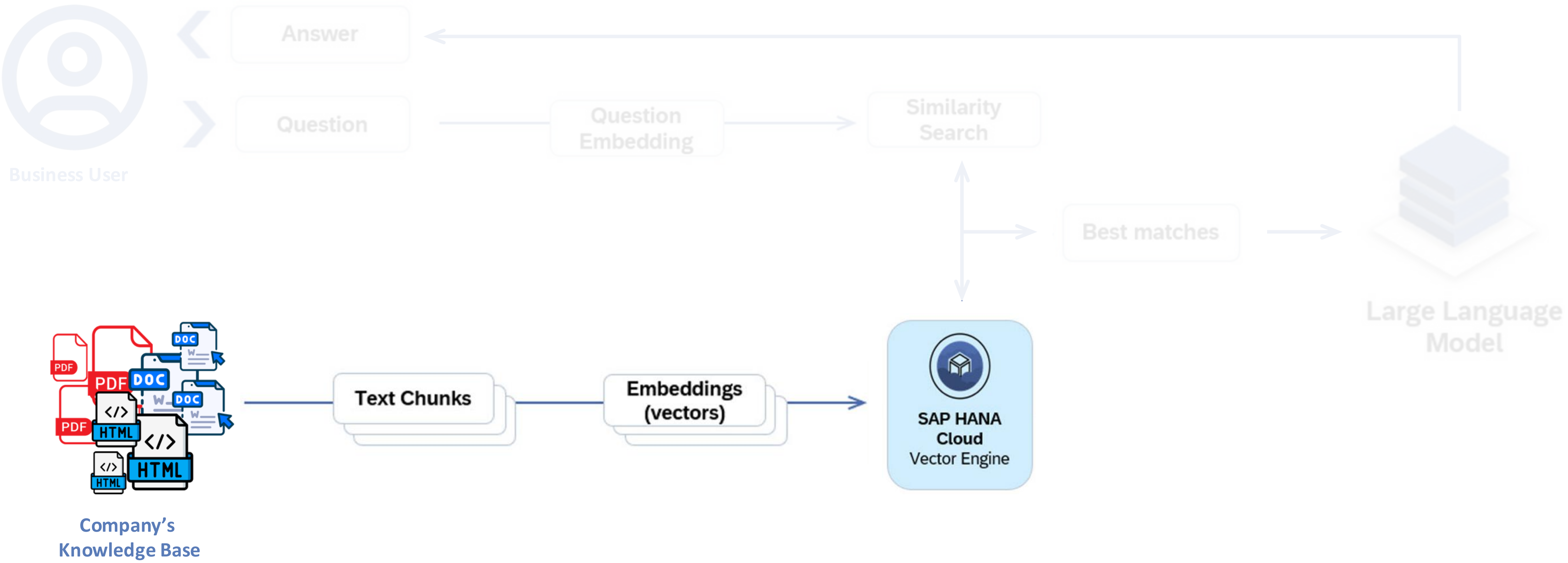


RAG – Retrieval Augmented Generation

Split the documents into text chunks

Generate the Embeddings from texts

Store the Embeddings



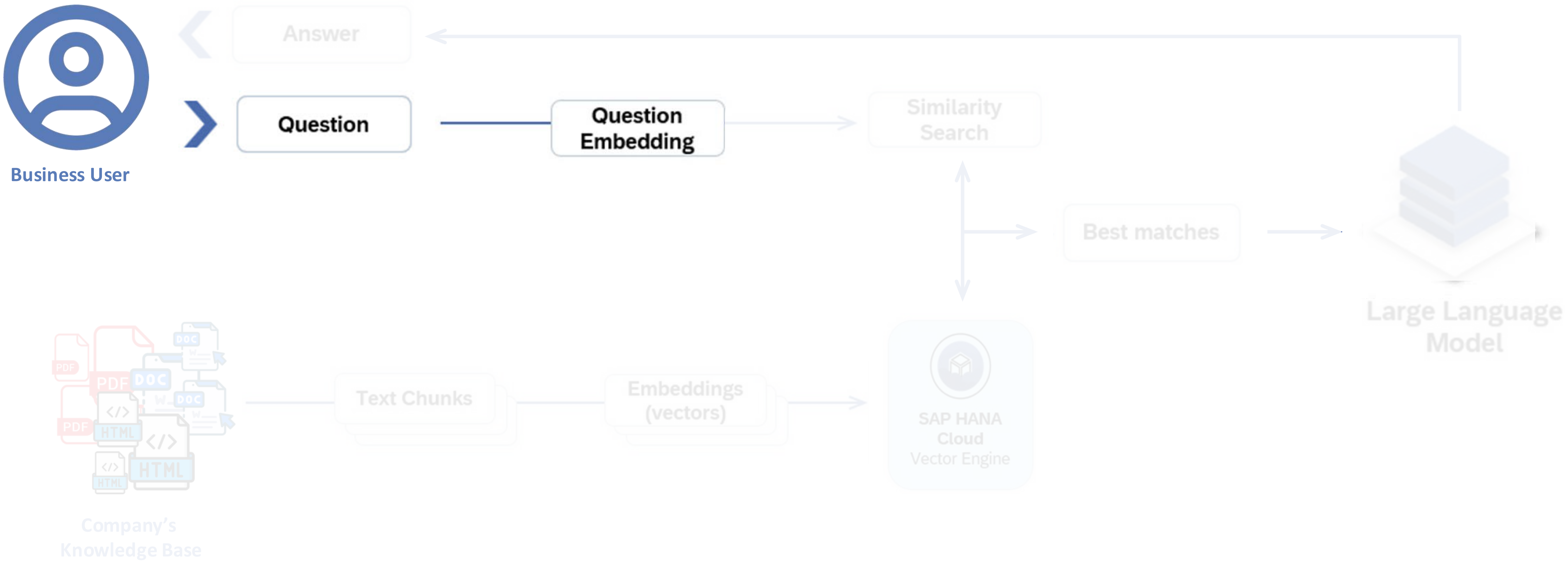
RAG – Retrieval Augmented Generation

Split the documents into text chunks

Generate the Embeddings from texts

Store the Embeddings

Generate Embeddings from the Question



RAG – Retrieval Augmented Generation

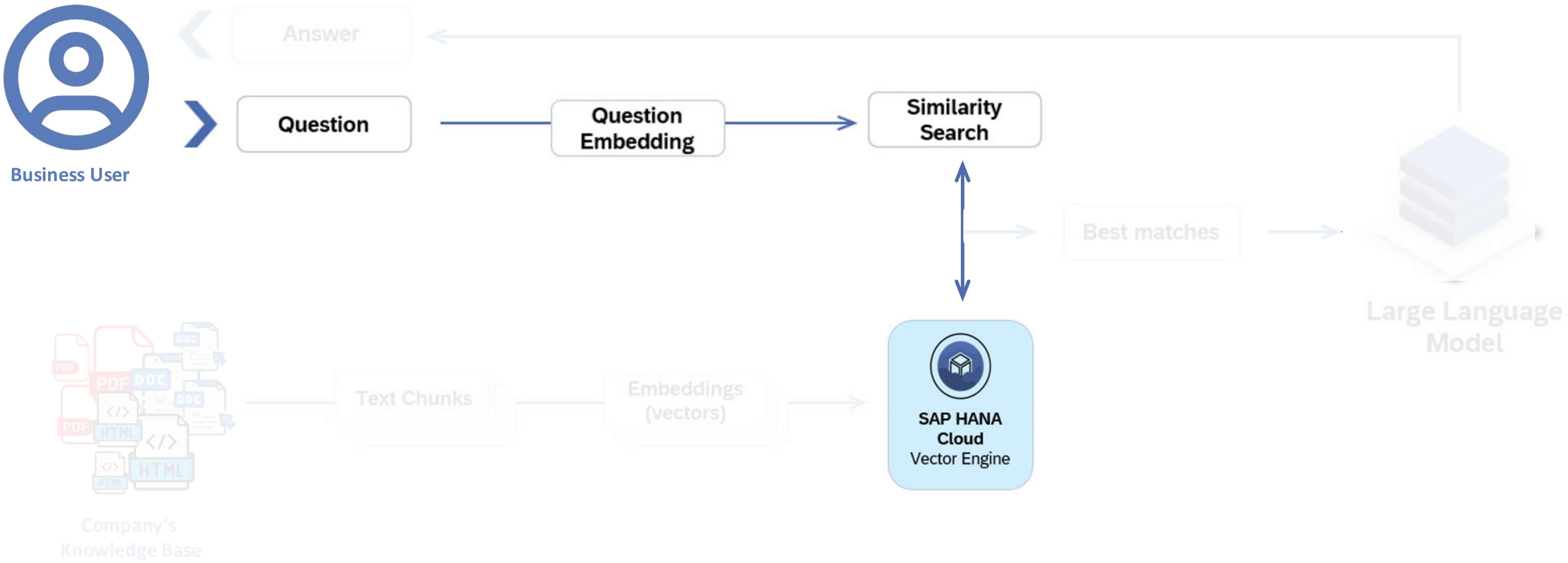
Split the documents into text chunks

Generate the Embeddings from texts

Store the Embeddings

Generate Embeddings from the Question

Get similar texts from Vector Engine



RAG – Retrieval Augmented Generation

Split the documents into text chunks

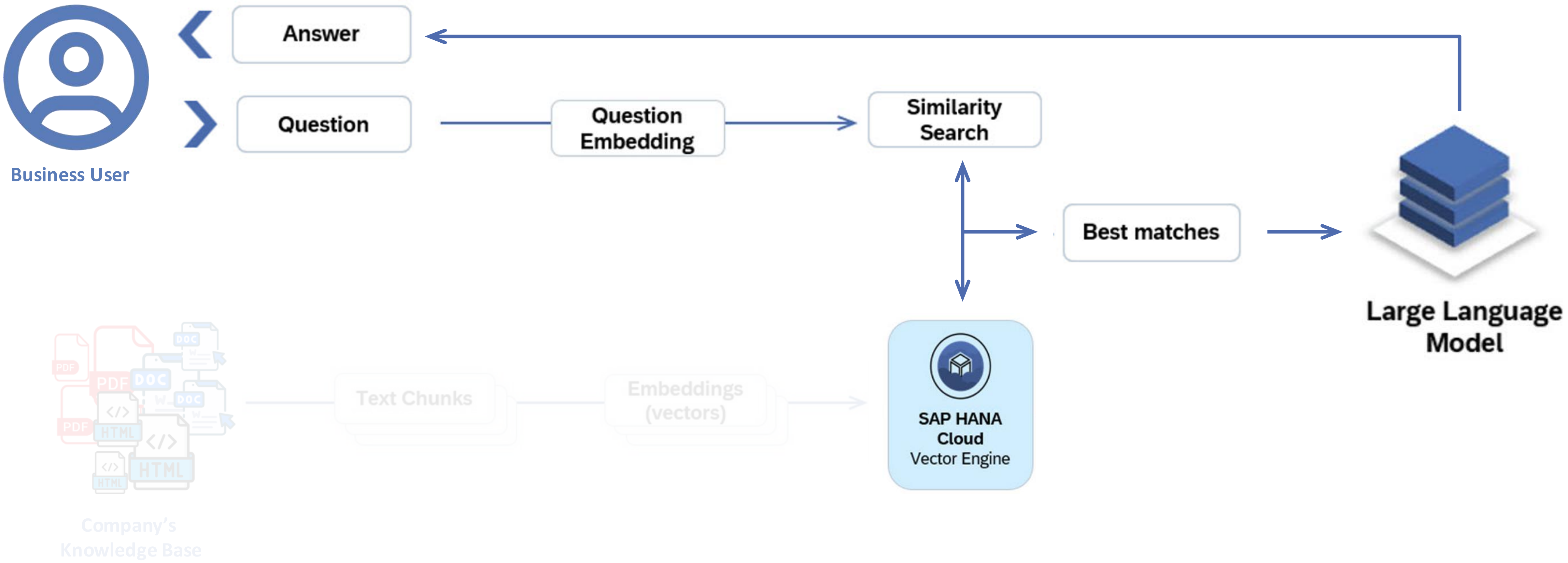
Generate the Embeddings from texts

Store the Embeddings

Generate Embeddings from the Question

Get similar texts from Vector Engine

Send **question** and **similar texts** to LLM



RAG – Retrieval Augmented Generation

Split the documents into text chunks

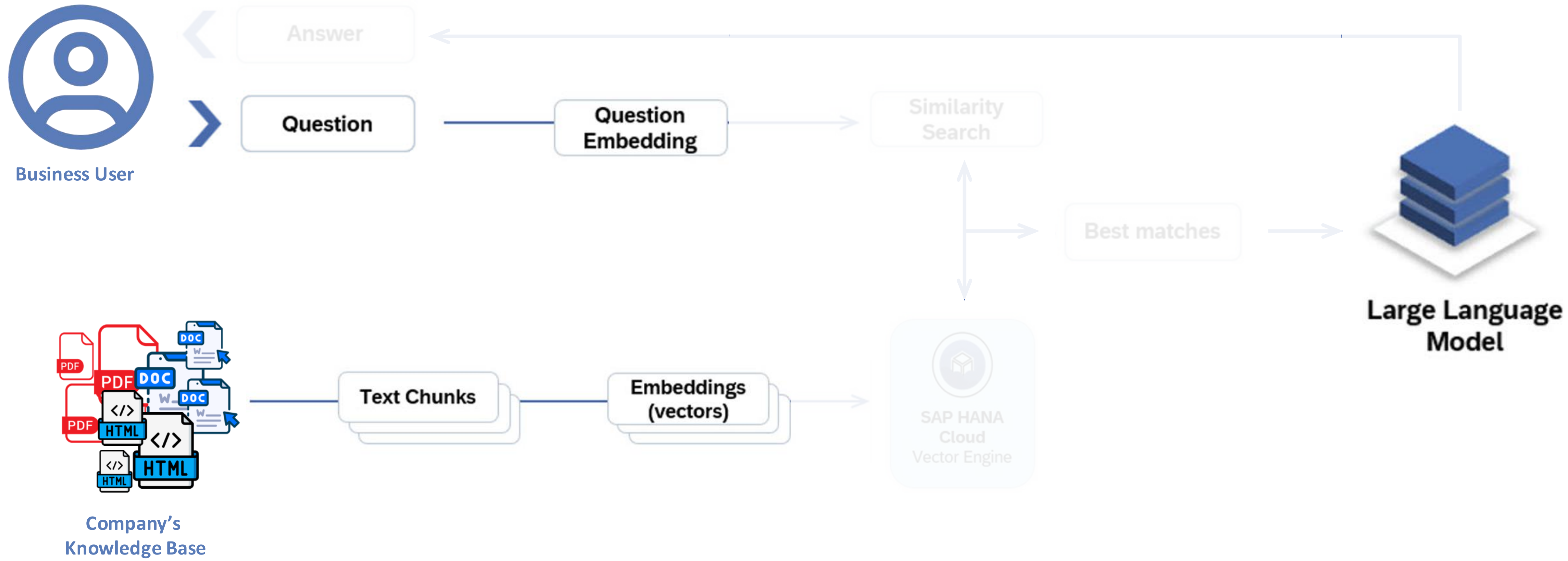
Generate the Embeddings from texts

Store the Embeddings

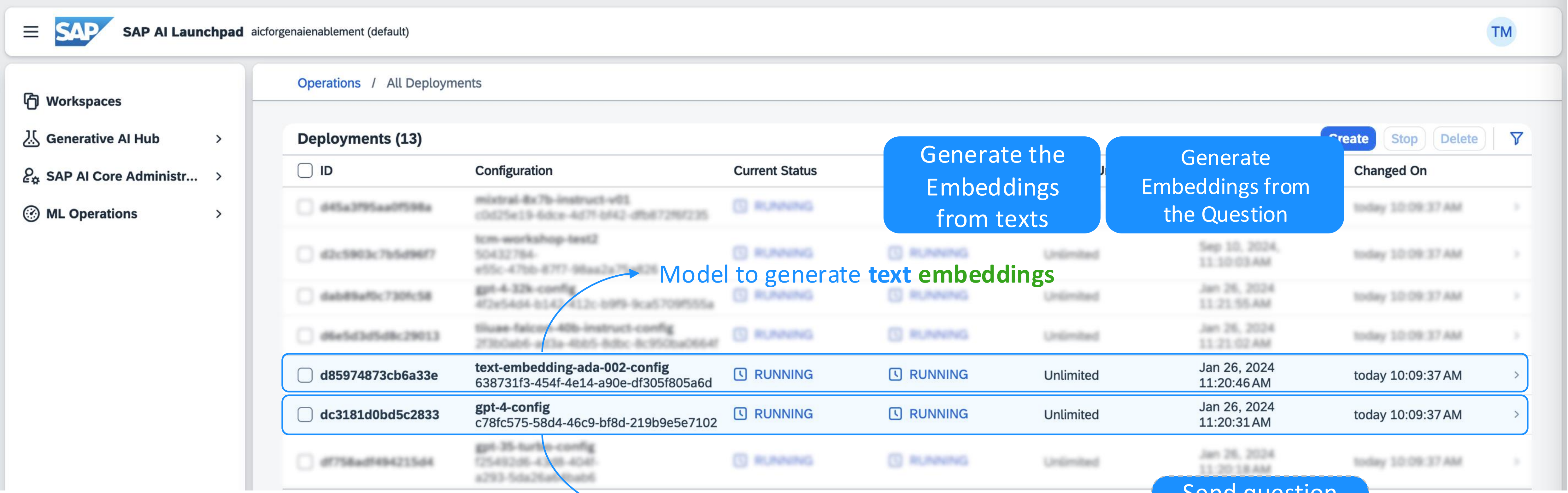
Generate Embeddings from the Question

Get similar texts from Vector Engine

Send question and similar texts to LLM



RAG – Retrieval Augmented Generation



Generate the Embeddings from texts

Generate Embeddings from the Question

Model to generate **text embeddings**

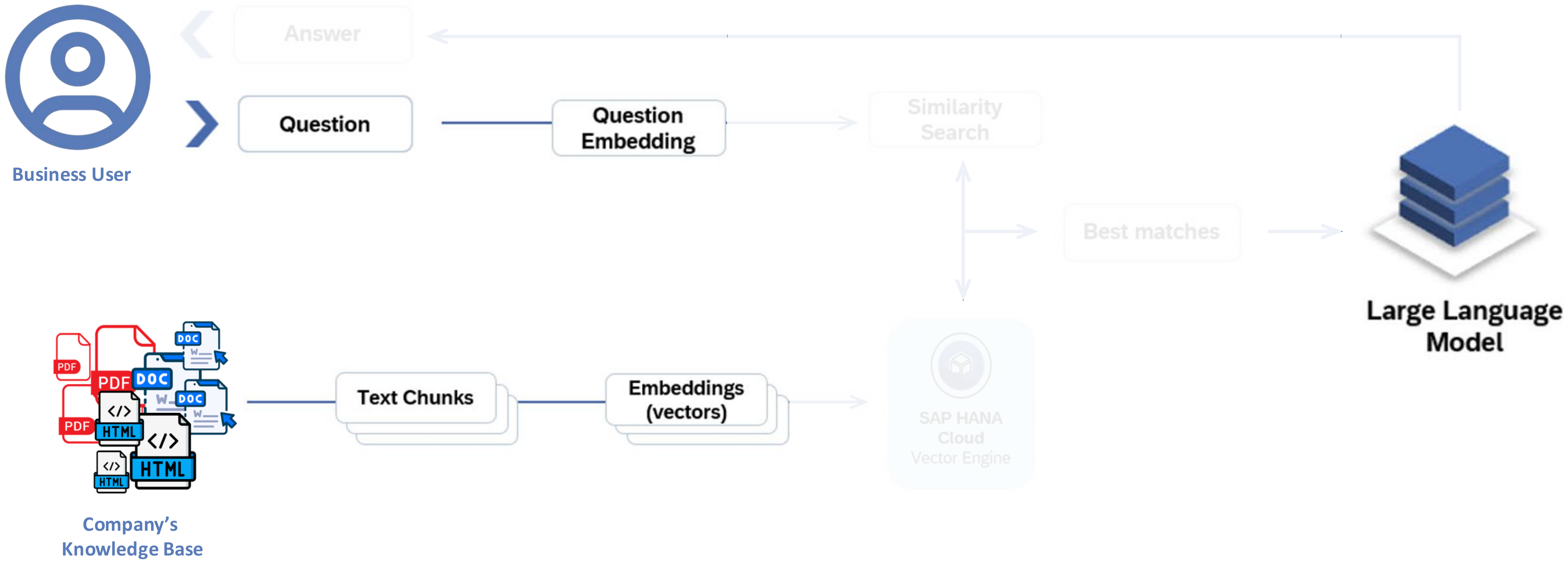
Model to generate **text completions**

Send question and similar texts to LLM

https://help.sap.com/docs/ai-launchpad/7837538aae8b4f6d832e018e2f499ea2/generative-ai-hub-in-sap-ai-launchpad

RAG – Retrieval Augmented Generation

- Split the documents into text chunks
- Generate the Embeddings from texts
- Store the Embeddings
- Generate Embeddings from the Question
- Get similar texts from Vector Engine
- Send question and similar texts to LLM



RAG – Retrieval Augmented Generation

Split the documents into text chunks

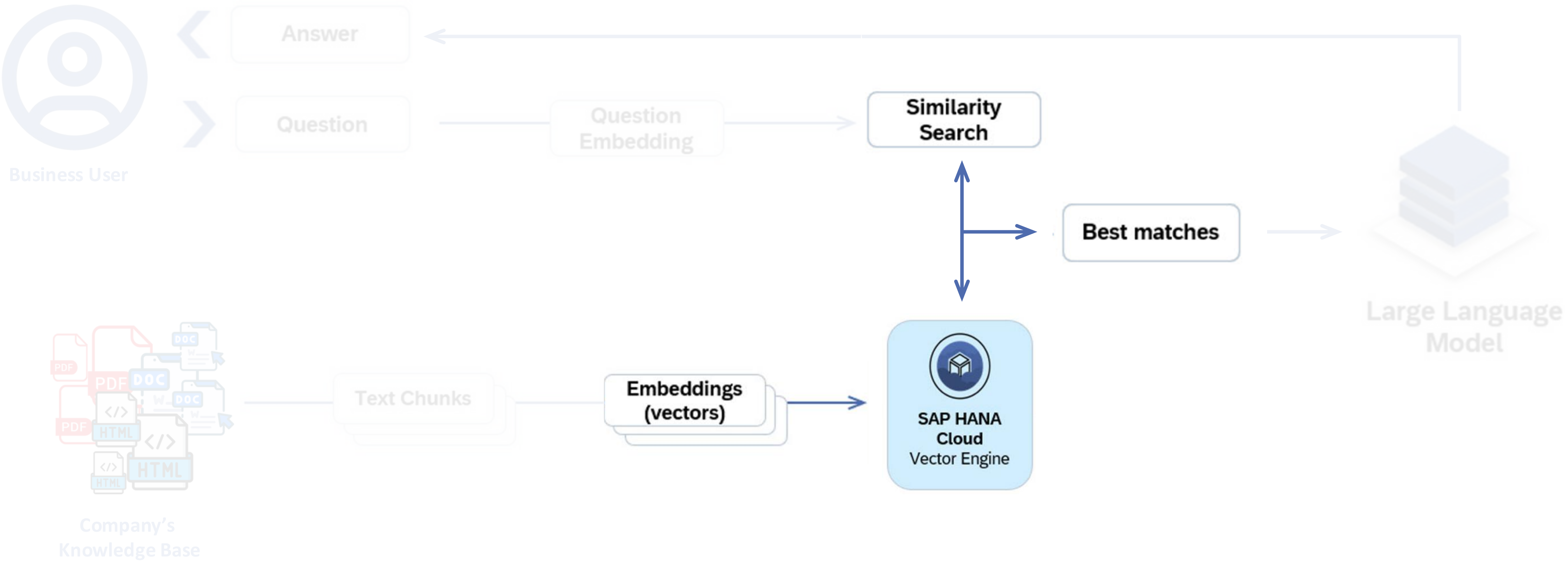
Generate the Embeddings from texts

Store the Vectors

Generate Embeddings from the Question

Get similar texts from Vector Engine

Send question and similar texts to LLM



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Similarity Search

Similarity search focuses on retrieving items that are similar or related to a given query, based on certain features or characteristics.

Examples:

- Content Recommendation Systems: Improving recommendations based on similar user preferences
- Image Retrieval: Searching for visually similar images in large datasets
- Product Matching: Matching similar products in e-commerce platforms

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Semantic Search

Semantic search involves understanding the meaning behind the user's query and the content being searched. This process goes beyond keyword matching and takes into account the context, intent, and relationships between words.

Examples:

- Natural Language Processing (NLP): Enabling more precise and context-aware language understanding
- Document Similarity Analysis: Identifying documents with similar semantic meaning
- Customer Support Chatbots: Enhancing conversational understanding for more accurate responses

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Information Retrieval

Information Retrieval (IR) is the process of obtaining relevant information from a large collection of data (such as documents, images, or web pages) in response to a user's query. It focuses on efficiently finding and ranking information that best satisfies the user's needs.

Examples:

- Web Search: Provide users with instant and relevant search results as they type

Thank you.